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Abstract: This contribution attempts to provide an updated draft text for draft new recommendation ITU-T M.rcem-AI "Requirements for AI-based customer experience management of telecom services" as found hereinafter. The reference text is [SG2-TD213-R1/PLEN \(2023-03\)](#).

Draft Recommendation ITU-T M.rcem-AI

Requirements for AI-based customer experience management of telecom services

Summary

This draft Recommendation provides the requirements for AI-based customer experience management ~~for customer experience smart management~~ of telecom services. The goal is to explain the use of AI technology to achieve closed-loop management of customer experience understanding, evaluation, analysis, guarantee and optimization from the OSS level. This draft Recommendation considers the application of AI technology in customer experience quality understanding, monitoring, analysis and optimization strategy generation.

Keywords

customer experience management, AI-based, Telecom Services.

Draft Recommendation ITU-T M.rcem-AI

Requirements for AI-based customer experience management of telecom services

1 Scope

This draft Recommendation provides the requirements for AI-based customer experience management of telecom services. The goal is to explain the use of AI technology to achieve closed-loop management of customer experience evaluation, analysis, guarantee and optimization from the OSS level. It mainly covers the following contents:

- 1) Functional requirements for AI-based customer experience management of telecom services.
- 2) Typical use cases for AI-based customer experience management of telecom services.

2 References

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

[ITU-T M.3080] Recommendation ITU-T M.3080 (2021), *Framework of AI enhanced Telecom Operation and Management (AITOM)*.

[TMF GB962] TMF GB962 (2017), *Customer Experience Management Solution Suite*.

[TBD]

3 Definitions

3.1 Terms defined elsewhere

This Recommendation uses the following terms defined elsewhere:

3.1.1 Customer experience management (CEM) [ITU-T E.800 series supplement 10]: An approach designed to focus on procedures and a methodology to satisfy the service quality needs of each end-user

[TBD]

3.2 Terms defined in this Recommendation

This Recommendation defines the following terms:

[TBD]

4 Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

AI Artificial Intelligence

AITOM AI-enhanced telecom operation management

CCR: Customer Churn Rate

CEM Customer experience management

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<u>CES:</u>	<u>Customer Effort Score</u>
CRM	Customer relationship management
<u>CRR:</u>	<u>Customer Retention Rate</u>
<u>CLV:</u>	<u>Customer Lifetime Value</u>
<u>CSAT</u>	<u>Customer Satisfaction</u>
<u>FCR</u>	<u>First Contact Resolution</u>
NMS	Network management system
<u>NPS:</u>	<u>Net Promoter Score</u>
OSS	Operation support systems
<u>SSA:</u>	<u>Self-Service Adoption</u>
[TBD]	

5 Conventions

In this Recommendation:

The keywords "is required to" indicate a requirement which must be strictly followed and from which no deviation is permitted, if conformance to this Recommendation is to be claimed.

[TBD]

6 Overview

The iterative evolution of the network has promoted the vigorous development of telecom business. Operators are highly concerned about how to improve the quality of customer experience, conduct objective and scientific analysis of customer experience data, make scientific, reasonable and rapid judgment on the quality of customer experience, accurately locate the factors affecting customer experience, and make the analysis data of customer experience quality create more reference value for production practice. Lifecycle management of the customer experience needs to be implemented from a service perspective.

Customer Experience Management (CEM) is the term used to describe a set of strategies that help nurture the relationships companies/service providers have with their customers to transform business and make it customer centric i.e. service provider exists to earn profit by delighting customer instead of defining strategy that is business centric where service provide runs business to earn profit and in journey satisfies customer. CEM transforming service management by providing an insight into customers' perceptions of services provided by telecoms operators/ service providers. Customer experience data has the following characteristics: rich variety, large quantity, and fast growth rate. AI-based optimization algorithms adapt to the needs of customer experience data analysis and dynamically reflect trends in service quality and key metrics. Customer experience management based on artificial intelligence can make network operators more efficient in network management and service.

The goal of the draft is to explain the requirements for AI-based customer experience management of telecom services from the OSS level, information such as monitoring and analysis results and diagnosis and optimization strategy is feed-back to network management system, work order system, customer service system and other telecommunication management systems to realize closed-loop management of customer experience evaluation, analysis, guarantee and optimization.

[TBD]

7 Framework for AI-based customer experience management of telecom services

7.1 Functional block diagram for AI-based customer experience management of telecom services

[Editor's note] This chapter will provide a functional block diagram for AI-based customer experience management of telecom services.

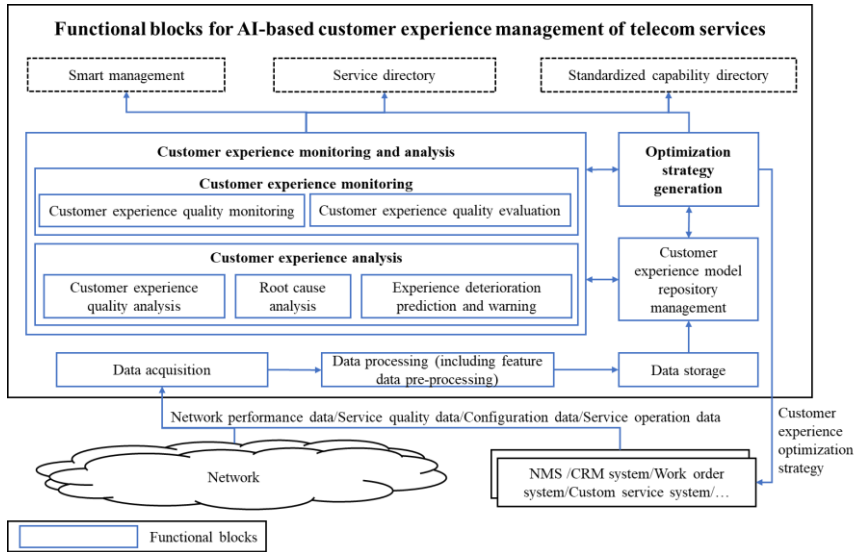


Figure 7-1 Functional block diagram for AI-based customer experience management of telecom services

NOTE1 – The blue blocks are the main functional blocks for AI-based customer experience management of telecom services.

NOTE2 – Network, NMS, CRM system, work order system, custom service system or other management systems can come from multiple manufacturers.

The functional block diagram of AI-based customer experience management of telecom services is shown in Figure 7-1, including:

- 1) Data acquisition: ~~This block~~ collects network performance data, service quality data, configuration data service operation data from network and NMS, CRM system, work order system, custom service system or other management system.
- 2) Data processing: The main functions of ~~this block data processing~~ refer to the chapter 8.3.3.2 of [ITU-T M.3041], including data cleaning and tagging and feature data pre-processing.
- 3) Data storage: ~~This block~~ storage the data from “Data processing-~~block~~”, and provides data for the “Customer experience model library management-~~block~~” to train AI model.
- 4) Customer experience model repository management: ~~This block~~ trains, storages and manages different service quality assurance models.
- 5) Customer experience monitoring and analysis: ~~This block~~ performs customer experience monitoring and analysis based on collected data and trained models, including customer experience quality monitoring, evaluation, analysis, root cause analysis and experience deterioration prediction and warning. The results of customer experience monitoring and

analysis will be sent to service directory and standardized capability directory and shown in smart management scenario application.

- 6) Optimization strategy generation: ~~This block~~ generates customer experience optimization strategy based on the results of customer experience monitoring and analysis. The results of customer experience optimization will be sent to service directory and standardized capability directory and shown in smart management scenario application.

NOTE3 – The functions of service directory, scenario application and standardized capability directory reference [ITU-T M.3080], and they are not be defined again in this Recommendation.

[TBD]

7.2 Relationship between framework for AI-based customer experience management of telecom services and AITOM framework

[Editor's note] This chapter will describe the relationship between framework for AI-based customer experience management of telecom services and AITOM framework.

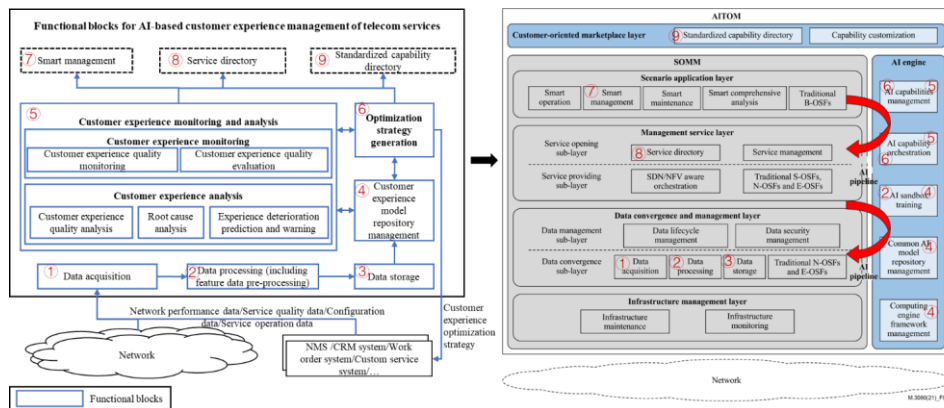


Figure 7-2 Mapping relationship between functional block diagram for AI-based customer experience management of telecom services and AITOM framework

NOTE1 - The symbols of red circle with number represent the mapping relationship between “Functional block diagram for AI-based customer experience management of telecom services and AITOM framework.

[TBD]

8. Functional requirements for AI-based customer experience management of telecom services

[Editor's note] This chapter will describe the Requirements for AI-based customer experience management of telecom services.

8.1 Data acquisition

The data includes network performance data, service quality data, configuration data, and service operation data from different sources. Data gathering is required to identify and prioritize areas of improvement and make adjustments accordingly

[TBD]

8.2 Data processing and feature data pre-processing

Data processing and feature data pre-processing can be realized based on AI algorithms.

- It is required to consider missing value handling, unit consistency, outlier handling and so on during data processing.
- It is required to consider feature data selection, feature data addition, normalization or standardization of feature data and feature extraction according to the specific application scenarios of customer experience quality monitoring, analysis and strategy generation.

[TBD]

8.3 Data storage

It is required to storage data from different sources to create a unified customer profile, allowing for a holistic view of each customer's interactions uniformly. Consistent data format, classified storage for different data and reasonable storage period are recommended to be considered.

[TBD]

8.4 Customer experience model repository management

Customer experience model repository management trains and management customer experience models.

[TBD]

8.5 Customer experience monitoring and analysis

It provides invaluable insights into how well different parts of CEM strategy are performing—and what changes need to be made in order to improve performance over time.

8.5.1 Customer experience monitoring

8.5.1.1 Customer experience quality monitoring

It is required to regularly monitor the quality of customer experience from multiple dimensions. Such as time dimension, regional dimension, business dimension, customer dimension, and so on.

[TBD]

8.5.1.2 Customer experience quality evaluation

Evaluate the quality of customer experience according to the monitoring results and threshold values.

[TBD]

8.5.2 Customer experience analysis

8.5.2.1 Customer experience quality analysis

It is necessary to analyze customer business characteristics, the strengths and weaknesses of customer experience and the changing trend of customer experience of single business and comprehensive business, as well as customer business preferences, so as to provide data for optimization strategy generation module and guarantee customer experience quality based on non-business characteristics.

[TBD]

8.5.2.2 Root cause analysis

Big data and machine learning technologies are used to analyze alarm data, resource data and topology data, and dynamic mining of alarm root cause analysis rules is realized, which supports rapid fault location and alarm convergence.

[TBD]

8.5.2.3 Experience deterioration prediction and warning

Artificial intelligence and big data technologies are applied to realize the prediction and early warning of customer experience deterioration and the dynamic adaptive optimization and adjustment of indicator thresholds.

Establish intelligent customer service capability, forecast and warn customer complaints, and care for key users in advance.

[TBD]

8.6 Optimization strategy generation

It is required to generate experience optimization strategies based on the analysis results of customer experience quality, and feed back to relevant systems for implementation to optimize customer experience.

- Dynamically adjust indicator thresholds.
- Sort the fault priorities based on service application scenarios.

NOTE - AI algorithms used in this block can be provided by AI Engine of AITOM framework in [ITU-T M.3080].

[TBD]

8.7 Ethical Considerations and Transparency

Data Privacy: Ensure compliance with data protection regulations and prioritize customer privacy.

Transparency: Maintain transparency in AI-driven processes, informing customers when AI is involved in interactions.

9. KPIs for AI-Based CEM

Key Performance Indicators (KPIs) for customer experience management based on AI help measure the effectiveness and impact of AI-powered initiatives on improving customer satisfaction and loyalty. Here are some essential KPIs to consider:

9.1 Net Promoter Score (NPS): NPS measures the customer loyalty and the likelihood of customers recommending the brand to others. Use a survey to ask customers how likely they are to recommend your brand to others on a scale of 1-10. The higher your customers rate you, the more likely they will promote your brand. AI can impact NPS by enhancing the overall customer experience and addressing customer issues promptly, leading to higher customer advocacy.

9.2 Customer Satisfaction Score (CSAT): Customer satisfaction scores indicate how satisfied your customers are with your service or product. CSAT scores are measured by creating surveys with predetermined scales for customers to rate their overall experience. AI can contribute to improving CSAT by providing quick and accurate responses to customer inquiries and offering personalized solutions.

9.3 Customer Effort Score (CES): Customer effort score reflects how much effort a customer puts into resolving an issue, completing tasks, or speaking with customer support.

9.4 Customer Churn Rate (CCR): Customer churn looks at how many customers you've lost over a specific period. It is more expensive and timely to obtain new customers than retain existing ones, so reducing churn should be a high priority. Tracking the churn rate allows businesses to assess the effectiveness of AI-powered retention strategies in reducing customer churn.

9.5 Customer Retention Rate (CRR): Customer retention rate measures how many customers a brand retains over time. High retention rates rely heavily on quality customer experience and customer service.

9.6 Customer Lifetime Value (CLV): Customer lifetime value measures how much money customers are likely to spend throughout their relationship with your brand.

9.7 First Contact Resolution (FCR) Rate: FCR measures the percentage of customer issues resolved during the first interaction. AI-powered virtual assistants and chatbots can significantly contribute to higher FCR rates by efficiently addressing customer queries and providing accurate solutions.

9.8 Self-Service Adoption (SSA) Rate: Measuring the percentage of customers who use self-service options powered by AI demonstrates the success of AI-driven self-help solutions and their impact on reducing the need for human intervention.

9.9 Customer Retention Cost vs. Acquisition Cost: Assessing the cost of retaining a customer compared to acquiring a new one helps understand the economic impact of AI-driven customer experience management and customer retention efforts.

Annex A

Typical use cases for customer experience management with AI

(This annex forms an integral part of this Recommendation.)

[TBD]

1. Predictive Customer Support: AI can predict potential customer issues based on historical data, allowing companies to offer proactive support. For example, an AI system can detect a potential network problem and notify affected customers before they even experience the issue.

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2. Fraud Detection and Security: Telecom providers handle vast amounts of sensitive data, making them an easy target for cyberattacks. Artificial Intelligence tools harnesses predictive analytics and end to end machine learning techniques to automatically identify anomalies by continuously learning from the past data. AI and machine learning algorithms analyze patterns and detect unusual behavior, identifying potential fraud or security breaches in real-time. Furthermore, with these telecom AI use cases, service providers can quickly respond to threats, safeguarding their infrastructure and customer data. AI's ability to learn and adapt to new fraud techniques makes it a capable and necessary tool in telecom security management.

3. AI-powered personalization: AI-powered personalization involves using artificial intelligence and machine learning to analyze customers' data, understand their preferences and needs, and tailor their experience accordingly. The implementation requires clear objectives, high-quality data, continuous testing and refinement, transparency, and integration across all touchpoints. The benefits of AI-powered personalization include improved customer experience, increased revenue, reduced churn, and data-driven insights.

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4. Chatbots And Virtual Assistants: Chatbots and virtual assistants turning everyday tasks into simple commands. We're already surrounded by these virtual assistants in the form of Amazon Alexa, Google Home, Apple's Siri and more. Many companies have created tasks with existing virtual assistants and chatbots so their customers can streamline their interactions with the company.

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5. Network Optimization: AI can analyze network traffic patterns and predict peak usage times, enabling telecom operators to optimize network resources for better call quality and data speeds during high-demand periods.

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Appendix I

The following table provides the gap analysis for the proposed new work item.

SDOs	Title	Analysis results
ITU-T SG2	Recommendation M.3080: Framework of artificial intelligence enhanced telecom operation and management (AITOM)	This Recommendation describes “Quality assurance capability maintenance” in AI capability orchestration. But it just gave a brief description without extending the content. This new work item can refer to AIOTM framework and focus on customer experience assurance or management.
ITU-T SG2	Recommendation ITU-T E.480: Framework for service management operational requirements – Service management	This Recommendation is intended to support and define the role of service management, and it provides a definition of service management and provides a framework for further analysis of the operational activities associated with service management. It does not include how to implement customer experience management based on AI models, including customer experience index model building, calculation, analysis, etc..
ITU-T SG2	Recommendation ITU-T M.3365: Requirements for quality of experience management of video used for surveillance	This Recommendation specifies, for video used for surveillance, requirements for QoE management, including that of resources, indicators, the evaluation activity configuration and evaluation records. It is based on single service experience management and does not use AI optimization algorithms.
ITU-T SG12	Recommendation ITU-T P.10/G.100: Vocabulary for performance, quality of service and quality of experience Recommendation ITU-T G.1011: Reference guide to quality of experience assessment methodologies Recommendation ITU-T E.803: Quality of service parameters for supporting service aspects Recommendation ITU-T Y.1541: Network performance objectives for IP-based services	SG12 focuses on the indicators and assessment methods of performance, quality of service (QoS), quality of experience (QoE) and classification of QoE assessment criteria, etc. These Recommendation focus on single business scenarios, such as IPTV, gaming, financial services, etc. This new proposed work item will define how telecom operators evaluate, monitor, analyse and manage the quality of customer experience of major telecom services with those performance, QoS, QoE, and how to apply AI technology to enhance customer experience.
TMF Customer Experience Management Project	GB962: Customer Experience Management Solution Suite GB1006: Customer_Experience_Management_2025_Guidebook_v2.0.1	TMF CEM project introduces the key imperatives for omnichannel implementation as well as a definition of the operational functions needed to successfully create an omnichannel experience for customers cross partner eco-systems. It also defines several

	IG1203: Capabilities for CEM Concepts and Processes v3.0.0 IG1303: Customer Experience Metrics v1.0.0-v8	typical business processes, such as establish customer engagement, manage content distribution to channels, etc. However, these processes are still abstract and high level, and cover many fields such as market, sales, products, services, resources, partners, etc. It does not specifically define how telecom operators evaluate, monitor, analyse and manage the quality of customer experience of major telecom services, nor does it involve how to apply AI technology to enhance customer experience in these areas. This new proposed work item will focus on the implementation of these contents.
3GPP SA4	TR 26.909 Study on improved streaming Quality of Experience (QoE) reporting in 3GPP services and networks	This recommendation studies the enhancement of Quality of Experience (QoE) for operator managed streaming service, 3 rd party managed streaming service and Over-The-Top (OTT) streaming service. It just focuses on the streaming service, and does not involve the implementation of customer experience management based on AI models, including customer experience index model building, calculation, analysis, etc.
3GPP SA4	TR 26.929 QoE parameters and metrics relevant to the Virtual Reality (VR) user experience	This recommendation studies key performance indicators that may impact the experience of VR service, identify and define new QoE parameters and metrics relevant to virtual reality user experience, which only involves the virtual reality user experience and the relevant QoE parameter. This new proposed work item focuses on how telecom operators evaluate, monitor, analyze and manage the quality of customer experience of major telecom services with performance, QoS, QoE, and how to apply AI technology to enhance customer experience.
3GPP SA5	TS28.404 Telecommunication management; Quality of experience(QoE) measurement collection; Concepts, use cases and requirement	This recommendation addresses concepts, business level use cases and requirements for the function Quality of Experience (QoE) measurement collection in 3GPP networks. The function includes collecting QoE information from UEs frequenting a specified area or an individual UE for a specified end user service/end user service type. It does not involve the implementation of customer experience management based on AI models using the QoE, including customer experience model building, calculation, analysis, etc.
3GPP RAN3	TR38.890 Study on NR QoE (Quality of Experience) management and optimizations for diverse services	This recommendation captures the results of study on NR QoE management and optimizations for diverse services. And it identifies to study the framework for

		triggering, configuring, collection and reporting of NR QoE measurement and study the potential impact on the related interface (e.g. F1, NG, Xn interface). It mainly focuses on the NR QoE management and optimization, does not involve the customer experience management of telecom services.
ETSI F5G	ETSI GS F5G 005 V1.1.1 (2022-03)	The present document studies the end-to-end Quality of Experience (QoE) factors for services over the broadband network. The key services discussed in the present document are typical Internet applications and Virtual Reality (VR). And AI modules deployed on edge network elements identify the different services. This new proposed work item will define how telecom operators evaluate, monitor, analyse and manage the quality of customer experience of major telecom services with those performance, QoS and QoE, and how to apply AI technology to enhance customer experience.
ETSI	ETSI TS 103 397 V1.1.2 (2020-03)	The present document provides guidance on OTT video streaming testing approach with a set of minimum desired and most meaningful QoE centric QoS parameters along with recommendations to create a figure of merit quantifying the OTT video streaming session quality, where possible. This new proposed work item will use performance, QoS, QoE, and apply AI technology to analyze, monitor and evaluate the customer experience of telecom services which the customer used.
ETSI	ETSI TR 103 702 V1.1.1 (2020-11)	The present document primarily concentrates on QoS parameters in relation to eMBB performance scenarios. And the scope of the document concentrates on measuring the network capability, assessed primarily through network QoS parameters, such as data rates, capacity, coverage, latency and continuity measurements. While the new proposed work item focus on the customer experience.
MEF	None	MEF focuses on solving metro Ethernet technology issues and network automation support services. The research object does not include the quality evaluation and customer experience of telecommunications services hosted on the network. There are no relevant criteria for searching using the keywords "experience", "quality", "QoS", "QoE".

IETF	None	IETF is only concerned with the calculation of related input parameters such as QOE/QOS indicators or the standard setting of the extension of related protocol messages such as IGP/BGP. It does not involve the research of CEM intelligent policy.
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